

# VEDANT KAPIL

AI Engineer | Generative AI & Full-Stack Systems

India | [Portfolio](#) | [GitHub](#) | [LinkedIn](#) | [Email](#)

## PROFESSIONAL SUMMARY

Impact-focused AI Engineer with proven hands-on experience architecting and deploying end-to-end Generative AI pipelines, Retrieval-Augmented Generation (RAG) architectures, and real-time Computer Vision models. Highly adept at building high-precision semantic search engines, agentic workflows, and local LLM pipelines using Python, LangChain, ChromaDB, and FastAPI. Experienced in integrating robust AI backends with high-performance React/Next.js client applications to deliver production-ready software solutions.

## TECHNICAL SKILLS

- **Programming & Core:** Python, C, C++, SQL (PostgreSQL/MySQL), TypeScript, JavaScript, HTML5/CSS3
- **Generative AI & LLMs:** LangChain, LangGraph, LangSmith, Autonomous AI Agents, RAG Architectures, Vector Embeddings, ChromaDB, Sentence Transformers, Ollama, Prompt Engineering
- **Machine Learning & Data:** Scikit-learn, Supervised & Unsupervised Learning, Feature Engineering, Exploratory Data Analysis (EDA), NumPy, Pandas, Matplotlib
- **Computer Vision:** YOLO (v8/v11), OpenCV, Roboflow, Video Stream Ingestion, Object Detection & Tracking
- **Backend & Web Systems:** FastAPI, RESTful APIs, Next.js, React, Tailwind CSS, JSON Schema
- **Developer Tools & Cloud:** Git, GitHub, Linux/Bash, VS Code, Vercel, Streamlit, Docker (Familiar)

## SELECTED TECHNICAL PROJECTS

### Personal Todo AI — Intelligent Task & Knowledge Platform | *Python, LangChain, ChromaDB, LLMs, Next.js, Vercel*

- Architected an intelligent task and knowledge management platform integrating RAG-based context retrieval over multi-format study notes, documents, and assignments.
- Engineered an automated document ingestion pipeline utilizing recursive chunking and Sentence Transformers, achieving low-latency vector similarity retrieval in ChromaDB.
- Implemented dynamic multi-model LLM routing (local & API-based) with customizable system prompts, cutting query latency while preserving context accuracy.

### Enterprise PDF RAG Intelligence Engine | *Python, LangChain, ChromaDB, Ollama, Streamlit*

- Engineered an end-to-end local RAG application enabling high-precision, context-grounded Q&A over complex multi-page PDF documents with zero data leakage.
- Configured vector embeddings via ChromaDB and optimized local inference using quantized models on Ollama, ensuring privacy-first on-device query resolution.
- Built an asynchronous Streamlit interface featuring token-by-token streaming responses and persistent conversation memory across multi-turn sessions.

### Real-Time Traffic Detection & Video Analytics System | *YOLO, OpenCV, Roboflow, Python*

- Developed a high-throughput computer vision pipeline using custom-trained YOLO models to detect, classify, and track multi-class vehicular traffic in real time.
- Curated, annotated, and augmented custom video datasets via Roboflow, substantially improving detection mAP under extreme lighting and occlusion conditions.
- Optimized frame-by-frame video stream parsing with OpenCV, achieving real-time inference at 30+ FPS with minimal CPU/GPU overhead.

### Production Client Web Application — Sindhu Construction | *Next.js, React, TypeScript, Tailwind CSS, Vercel*

- Architected and deployed a full-stack, responsive commercial web platform for a Bengaluru-based construction firm, driving organic discovery with a 95+ Google Lighthouse score.
- Developed dynamic client-facing components including an interactive project monograph, before/after renovation showcase, and an instant cost-estimator.
- Implemented robust SEO schema markup, metadata configurations, and CDN asset caching on Vercel to optimize core web vitals and organic traffic.

## Automated Dynamic Developer Portfolio & Headless CMS | *React, TypeScript, GitHub REST API, Tailwind CSS, GitHub Pages*

- Designed an automated developer portfolio integrating GitHub REST APIs to sync live repository metrics, commit activities, and project logs without manual updates.
- Built a lightweight administrative interface utilizing secure Personal Access Token authentication for remote content and state management.

## HACKATHONS & ACHIEVEMENTS

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### Code War Hackathon — AI-Powered Smart Cart & Automated Billing System | *5th Place*

- Developed a computer vision-based smart shopping cart using object detection to identify products in real time.
- Implemented automated invoice generation based on detected products, reducing manual checkout effort.
- Integrated the object detection pipeline with inventory and billing logic to create an end-to-end automated shopping experience.
- Presented and demonstrated the solution to judges during the hackathon.

## EDUCATION

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**Bachelor of Engineering in Computer Engineering** | St. John College of Engineering and Management *(2024 – 2028)*

**Higher Secondary Certificate (Class XII) – Science (PCMB)** | Motherland Public School (CBSE) — 80% *(2022)*

**Secondary School Examination (Class X)** | Motherland Public School (CBSE) — 76% *(2020)*

## CERTIFICATIONS & VIRTUAL EXPERIENCE

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- **Machine Learning Specialization** — DeepLearning.AI & Stanford University (Coursera)
- **Introduction to Large Language Models & Google AI Essentials** — Google Cloud / Google
- **CS50's Introduction to Programming with Python** — Harvard University
- **OpenAI Academy: Agents and Workflows** — OpenAI
- **Quantitative Research & Data Analytics Virtual Experience** — JPMorgan Chase & Co., Deloitte, Tata (Forage)